

Chapter 3 - What is DNS

1. DNS (Domain Name System)

What is DNS?

DNS is a system that **converts domain names (google.com) into IP addresses (142.250.195.78)**.

Humans remember names, computers understand numbers → DNS bridges the gap.

2. How DNS Works (Step-by-Step)

When you type **google.com** in a browser:

Step 1: DNS Resolver

- Your device sends the request to a **DNS Resolver** (usually provided by your ISP or Google DNS 8.8.8.8).
 - The resolver tries to find the IP address.
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Step 2: Check Cache

- Resolver checks if the IP is already stored in **DNS cache**.
 - If yes → returns instantly (fast response).
 - That's why **DNS cache should be cleared regularly** to remove outdated data.
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Step 3: Root Server

- If not cached, resolver contacts a **Root DNS Server**.
- Root servers know where TLD servers are located.

There are only **13 sets** of root servers worldwide (A–M).

Step 4: TLD Server (Top Level Domain Server)

- Handles domains like:
 - .com
 - .in
 - .org
 - .net
 - TLD tells the resolver which **Authoritative DNS server** holds the actual record.
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Step 5: Authoritative DNS Server

- Stores final DNS records.
 - Example: Google's DNS server returns the actual IP of google.com.
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Step 6: IP Sent to Client

- Resolver sends IP back to your browser.
 - Browser connects to that server.
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Why DNS Is Used?

- Humans cannot remember IP addresses.
 - Helps load websites faster.
 - Organizes the internet naming system.
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Daily-Life Example

- Typing **youtube.com** instead of its complex IP address.
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3. Public DNS Servers

Google DNS:

- **8.8.8.8** (most famous open DNS)

Cloudflare DNS:

- **1.1.1.1**
 - Faster than Google DNS in many regions.
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4. Binary to Decimal Conversion

Each octet of an IP is **8 bits**, so maximum value = **255**
(11111111 in binary)

Why it cannot exceed 255?

Because 8 bits can represent values from **0 to 255** only.

5. Ping (Packet Internet Groper)

What is Ping?

- A tool used to **check connectivity** between your device and a server.

How it works?

- Sends an ICMP echo request.
- If the server replies → connection is alive.

Why used?

- To check delay, connectivity issues, or packet loss.

Example

- ping google.com
Used by network engineers daily.
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6. IPv4 Limitations

$2^{32} = \sim 4.29$ Billion IPs

Total IPv4 addresses ≈ 4.29 billion.

Major Authorities:

- **IANA (Internet Assigned Numbers Authority)**
→ Manages global IP ranges.
- **RIR (Regional Internet Registries)**
→ Assign IPs to countries and ISPs.

Problem:

Even these organizations couldn't manage IP shortage →
→ So IP **Classes** were created.



1 CHAPTER SUMMARY



1. What is DNS?

DNS (Domain Name System) is a system that converts **human-readable domain names** into **machine-readable IP addresses**.



Humans remember names → Computers understand numbers



DNS acts as a **translator / bridge**



Example:

google.com → 142.250.195.78



2. How DNS Works (Step-by-Step)

When you type **google.com** in your browser  



Step 1: DNS Resolver

- Your system sends request to a **DNS Resolver**
- Usually provided by:
 - ISP
 - Google DNS → **8.8.8.8**



Resolver's job: *Find the correct IP address*



Step 2: DNS Cache Check

- Resolver checks its **cache**
- If IP exists → instant reply ⚡
- If outdated → problem occurs



That's why **clearing DNS cache** is important

● Step 3: Root DNS Server

- If not cached, resolver contacts **Root DNS Server**
- Root servers know **where TLD servers are**

🌐 Only **13 sets of root servers (A–M)** exist worldwide

● Step 4: TLD Server (Top Level Domain)

Handles domains like:

- .com
- .in
- .org
- .net

📌 TLD server tells:

➡ *Which Authoritative DNS server has the final record*

● Step 5: Authoritative DNS Server

- Stores **actual DNS records**
- Returns the **final IP address**

📌 Example: Google's authoritative DNS gives IP of google.com

● Step 6: IP Returned to Client

- Resolver sends IP to browser
 - Browser connects to the server
 - Website loads 🎉
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? Why DNS Is Used?

- ✓ Humans can't remember IPs
- ✓ Faster website loading
- ✓ Organized Internet naming system

 Daily Life Example:

Typing **youtube.com** instead of complex IP numbers 🤖

🌐 3. Public DNS Servers

🔵 Google DNS

- **8.8.8.8**
- Reliable & widely used

🔵 Cloudflare DNS

- **1.1.1.1**
 - Faster & privacy-focused 🚀
-

4. Binary to Decimal Conversion (IP Basics)

- Each IP octet = **8 bits**
 - Maximum value of 8 bits:
 - $11111111 = 255$

📌 So IP range per octet = **0 – 255**

? Why not more than 255?

👉 Because **8 bits = 256 values (0–255)**

5. Ping (Packet Internet Groper)

🔍 What is Ping?

A network tool used to **check connectivity** between your device and a server.

How Ping Works

- Sends **ICMP Echo Request**
- Server replies with **Echo Reply**
- If reply received → connection alive 

Why Ping Is Used?

- ✓ Check network connectivity
- ✓ Measure delay (latency)
- ✓ Detect packet loss

 Example:

ping google.com

 Used daily by network engineers

6. IPv4 Limitations

IPv4 Address Space

- IPv4 uses **32 bits**
- Total IPs:

$2^{32} \approx 4.29 \text{ Billion}$

Major IP Authorities

◆ IANA

- Internet Assigned Numbers Authority
- Manages global IP address pool

◆ RIR

- Regional Internet Registries

- Allocate IPs to countries & ISPs
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✗ Problem: IP Shortage

Even IANA & RIR couldn't handle demand 🤖

➡ Result:

- IPv4 exhaustion
 - Introduction of **IP Classes** (temporary solution)
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CONCLUSION

DNS is the **phonebook of the Internet** 📖 🌐

Without DNS:

- ✗ Websites won't load
- ✗ Internet becomes unusable

This chapter builds:

- ✓ Core networking knowledge
- ✓ Cyber Security fundamentals
- ✓ Real-world troubleshooting skills

👉 **DNS knowledge is mandatory for Networking, Cyber Security & Cloud roles** 📁 🔥



3 MIND MAP

Lecture 3 – DNS

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| — DNS

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| — Domain → IP

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| — Human ↔ Computer bridge

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| — DNS Working

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| — Resolver

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| — Cache

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| — Root Server

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| — TLD Server

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| — Authoritative Server

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| — IP to Client

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| — Public DNS

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| — Google → 8.8.8.8

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| — Cloudflare → 1.1.1.1

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| — IP Basics

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| — Binary

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| — Max value → 255

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| — Ping

|

| — ICMP

| └─ Connectivity

| └─ Latency

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└─ IPv4 Limitation

└─ 2^{32} addresses

└─ IANA

└─ RIR

└─ IP shortage





Q & A

? Q1. What is DNS?

✓ Answer:

DNS converts domain names into IP addresses so computers can locate servers easily.

? Q2. Why DNS is required?

✓ Answer:

Because humans remember names while computers communicate using IP addresses.

? Q3. Explain DNS working?

✓ Answer:

Client → Resolver → Cache → Root → TLD → Authoritative → IP returned.

? Q4. What is DNS Resolver?

✓ Answer:

A server that finds the IP address for a domain name on behalf of the client.

? Q5. What is DNS Cache?

✓ Answer:

Temporary storage of DNS records to improve speed and reduce load.

? Q6. What are Root DNS Servers?

✓ Answer:

They direct queries to appropriate TLD servers. There are 13 root server sets worldwide.

? Q7. What is TLD?

✓ Answer:

Top Level Domain like .com, .org, .in.

? Q8. What is an Authoritative DNS Server?

✓ Answer:

It stores the final and correct DNS records of a domain.

? Q9. What is Google DNS?

✓ Answer:

A public DNS server with IP **8.8.8.8**.

? Q10. What is Cloudflare DNS?

✓ Answer:

A fast and privacy-focused public DNS server **1.1.1.1**.

? Q11. Why IP octet max value is 255?

✓ Answer:

Because 8 bits can represent values from 0 to 255 only.

? Q12. What is Ping?

✓ Answer:

A tool that checks connectivity using ICMP echo requests.

? Q13. Which protocol does Ping use?

✓ Answer:

ICMP (Internet Control Message Protocol).

? Q14. What does Ping check?

✓ Answer:

Connectivity, latency, and packet loss.

? Q15. What is IPv4?

✓ Answer:

A 32-bit IP addressing system with around 4.29 billion addresses.

? Q16. Why IPv4 addresses are limited?

✓ Answer:

Because 32-bit addressing allows only 2^{32} combinations.

? Q17. What is IANA?

✓ Answer:

An organization that manages global IP address allocation.

? Q18. What is RIR?

✓ Answer:

Organizations that distribute IP addresses region-wise.

? Q19. What problem occurred due to IPv4?

✓ Answer:

IP address exhaustion.

? Q20. How DNS is important for Cyber Security?

✓ Answer:

DNS is used in phishing detection, malware control, and traffic filtering.

KNM